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Requirements and models you can trust

Lecture 2 · CADE30008 Flight Dynamics & Advanced Control

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How much of a problem is this?

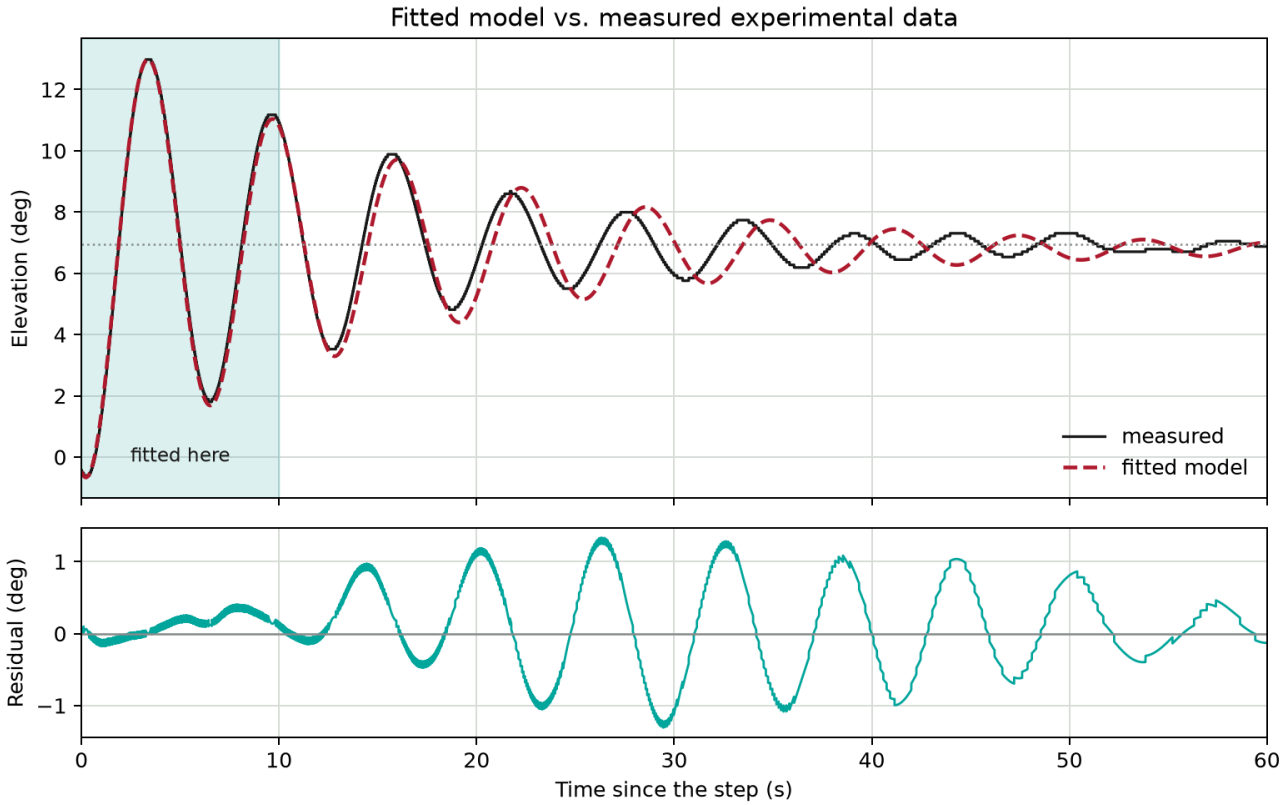
Hook and recall · 5 min

Before anything new

1. You fitted K, ω_n and ζ from one recording. Which do you trust **most**, and which **least**?
2. A classmate's ζ differs from yours by 10%. Is one of you wrong?
3. Your design met the requirement in simulation. What would have to be true for it to meet it on the rig?

Answer all three before any answer goes up.

The model, on top of the recording it came from



How much of a problem is this?

It depends what you want it for

Over the first	They differ by	Worst case
10 s	**2.8%** of the 6.80 deg step	0.41 deg
60 s	**8.9%**	1.33 deg

The model did not get worse. It was always a claim about a window.

One recording, one trim, one step size

- The arm was already swinging, through **4.21 deg**, when the step arrived
- `L02_validate` fits the starting state, which is the shaded window. So does MATLAB's `compare`
- Two minutes with `plot_recording` in the room would have saved it needing the repair
- Even from rest, a fit is a claim about **one trim elevation and one amplitude**

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What do we want?

Learn A · 15 min

A number, a test, and a reason

Where the number came from	Example
The vehicle's job	A camera swath sets the tracking tolerance
A handling-qualities criterion	Closed-loop damping inside the standard's band
A physical limit you measured	The demand saturates, so cap the controller
An agreement in the room	"Eight seconds", for want of a reason

The last one is legitimate. Dressing it up as one of the first three is not.

A settling time is a frequency

$$\zeta = \frac{-\ln M_{\mathrm{p}}}{\sqrt{\pi^2 + \ln^2 M_{\mathrm{p}}}}, \qquad \omega_{\mathrm{n}} = \frac{4}{\zeta\, t_{\mathrm{s}}}$$

Settle to 2% in	Overshoot at most	ζ	ω_n
1.5 s	10%	0.591	4.51 rad/s
2.0 s	10%	0.591	3.38 rad/s
3.0 s	10%	0.591	2.26 rad/s
4.0 s	20%	0.456	2.19 rad/s

Your model has to be right at that frequency.

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Make these testable

Do A · 10 min

YOU DO

Three things somebody actually said

1. "It should feel crisp"
2. "It mustn't wander off while the camera is running"
3. "Don't let it fight the pilot"

> Rewrite each as a requirement. Name **which of the four sources** its number came from, and **one number you invented**.
>
> **In pairs, ten minutes.** Done when all three have a quantity and a number.

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What do we know?

Learn B · 15 min

A model is trustworthy when we know three things

- **Over what frequencies, or over what time.** 10 s, not 60 s
- **Over what amplitude, and about what trim**
- **How far out it is inside that**, from a residual

> **Not being right. Nothing is right.** A model handed over without these three is three coefficients and a hope.

The residual's shape is the diagnosis

What it looks like	What it is telling you
Flat, the size of the sensor noise	Nothing systematic left
A slow drift one way	The gain is out, or the trim moved
Oscillation at the signal's frequency, growing	The **frequency** is out
Oscillation at some **other** frequency	A **missing mode**

None of this is visible in a fit percentage.

A fit percentage hides a great deal

- **Fitted over what?** Match the starting state over 60 s instead of 10 s: the 60 s residual improves from 8.9% to 8.3%, and the 10 s residual doubles, 2.8% to 6.7%
- **Against what?** 5% of the change the input asked for, or 5% of the signal's own range, are different numbers

> Least squares spreads the error over whatever you hand it.

"Near the trim" is a number

A linear model cannot, whatever its coefficients:

- **change its gain** with the size of the input
- **change its natural frequency** with the size of the motion

> So a measured dependence on amplitude is **outside the model by construction**, and no refit brings it inside.

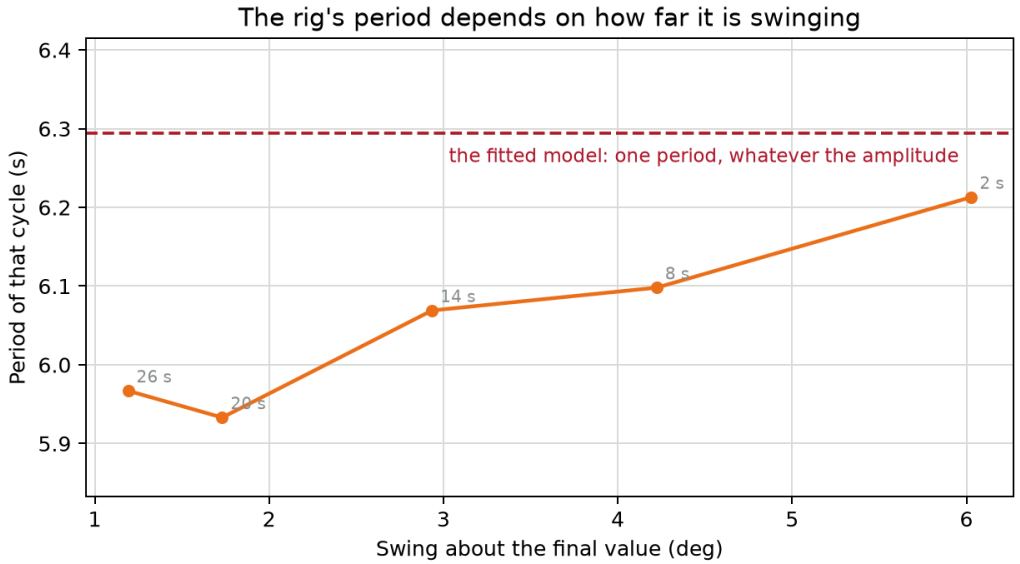
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Where does it fit?

Do B · 10 min

YOU DO

Measure the period, one cycle at a time



`L02_validate.m` · from **6.213 s** at **6.03 deg** of swing to **5.967 s** at **1.19 deg**

In pairs, ten minutes. Done when you can say the band of amplitudes your model is good for.

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The case

Case brief · 5 min

A pitch model, and two flight-test records

$$G(s) = \frac{\theta(s)}{\delta_{\mathrm{e}}(s)} = \frac{40}{s\,(s+2)\,(s+10)}$$

- Derived from geometry and aerodynamic data, and **handed to you**
- Two records, 60 s each, logged at 50 Hz: elevator, pitch rate, attitude
- Each flies a **3-2-1-1**, then a **pulse held and released**
- One at **2 degrees** of elevator, one at **8**

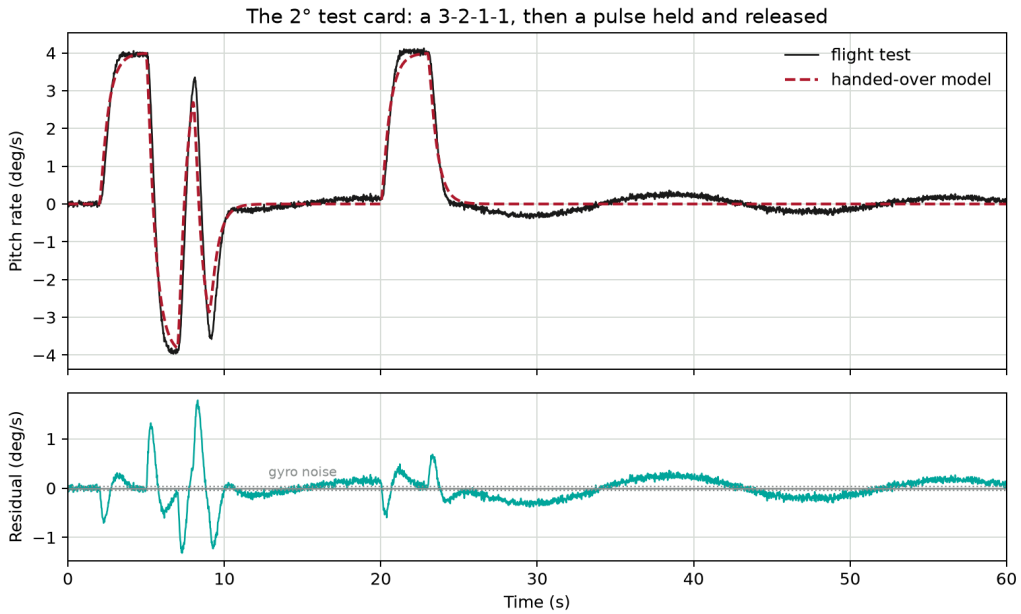
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Validate it, bound it, write the requirement

Case build · 20 min

Core expectation: simulate, subtract, look

YOU DO



Residual **14.2%** over the 3-2-1-1, **4.5%** after the release · gyro noise **1.0%**

Pairs or threes, twenty minutes. Done when you have the overlay, the residual, and the three observations.

Observation 1: after the release, the record waves and the model is flat

- From 24 s the elevator is back at zero, and the model's rate decays to zero with it
- The record oscillates anyway: period **17.9 s**, so **0.35 rad/s**, about **0.28 deg/s**
- Two real poles and an integrator **cannot oscillate at any frequency**

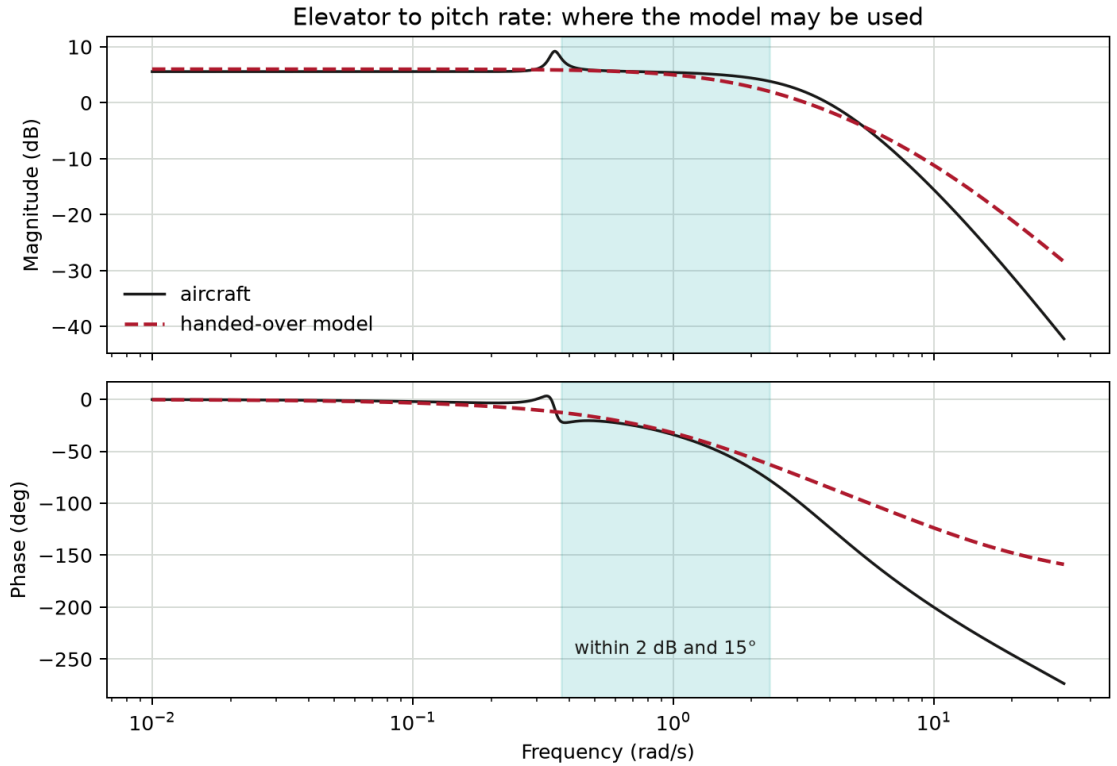
> **A residual with a frequency in it is a mode the model does not have.**

Observation 2: not wrong, wrong above a frequency

Everything below	2 deg record	8 deg record
0.5 rad/s	1.0%	1.8%
1.0 rad/s	1.8%	2.8%
2.0 rad/s	5.1%	4.6%
3.0 rad/s	8.5%	7.1%
10 rad/s	13.8%	12.0%

That is a sentence a designer can act on.

What the aircraft actually was



Agreement within 2 dB and 15° from **0.37** to **2.35 rad/s**

Observation 3: one gain in the model, two in the data

Elevator	Pitch rate per degree
2 degrees	**1.980** (deg/s)/deg
8 degrees	**1.798** (deg/s)/deg
The model, at any amplitude	2.000 (deg/s)/deg

A surface at a large deflection is less effective than the same surface at a small one.

Extension: can I ask my model to do this?

- **Settle to 2% in 3.0 s**, overshoot at most 10% → ω_n = **2.26 rad/s**, inside the band
- **Settle to 2% in 2.0 s → 3.38 rad/s**, outside it

> Relax the requirement, or go and fly more test points. **Designing there anyway and reporting the simulation is not an option.**

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What the room found

Converge · 10 min

Three observations and one argument

- **The bound.** Between the slow mode and about 2 rad/s, a few degrees of elevator, around 5% inside that
- **The missing mode**, at 0.35 rad/s
- **The amplitude.** One gain in the model, two in the data

> **Is this model good enough?** It depends entirely on the requirement.

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What this cannot do

Cliffhanger · 10 min

You have both halves, and it still does not work

- A model you can defend, with its range written down
- A requirement that can be met, missed or argued about
- A linear analysis that is **correct**

And the response goes a long way past where it was asked to stop.

Today, in five lines

- A model is a **claim with a range of validity**: frequencies, amplitude, and how far out inside that
- **Read the residual's shape**, not its size
- **Quote the window.** Same model, same recording: 2.8% over 10 s, 8.9% over 60 s
- **Linearisation is about a trim and an amplitude.** The rig's period moved 4%; the aircraft's gain moved 9%
- A requirement names a quantity, a number, a test **and a source**

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A model you trust and a requirement you can test

How to use them properly?

See you next week